

Qualification Pack



Cloud Computing Associate

QP Code: NIE/ITS/Q1203

Version: 1.0

NSQF Level: 4.5

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NIE/ITS/Q1203: Cloud Computing Associate

Brief Job Description

It intends to build a sound Cloud Computing professional for the industry. Based on this belief the concepts of Docker and Cloud Security have been added into this module in order to enable the enrolled probationer to get abreast with the concept of containerization in which the writing of code and its deployment into production environment has been fine-tuned. Cyber security is the need of the hour and plays all the more a critical role in the cloud computing environment

Personal Attributes

It include problem-solving, adaptability, and analytical thinking, which are essential for resolving issues and staying current with evolving technologies. Attention to detail and effective communication are crucial for managing cloud resources and collaborating with colleagues and clients. Strong time management and teamwork skills help in efficiently handling multiple tasks and working effectively in project teams

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ITS/N1217: Conceptualize Cloud Security](#)
2. [NIE/ITS/N1216: Conceptualize Docker and Containers](#)
3. [NIE/ITS/N1215: Emphasizing Cloud Computing Services](#)
4. [NIE/ITS/N1214: Managing Cloud Architecture with Open Stack](#)
5. [NIE/ITS/N1213: Conceptualize Advance Linux](#)
6. [NIE/ITS/N1212: Implementing Virtualization and Basics of Cloud Computing](#)
7. [NIE/ITS/N1211: Introduction to Network Management](#)
8. [NIE/ITS/N1210: Conceptualize Operating Systems \(Windows and Linux\)](#)
9. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization

Qualification Pack

Occupation	Cloud Management
Country	India
NSQF Level	4.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	2511.9900
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) with NA of experience OR Pursuing 1st year of UG (and continuous education) with NA of experience OR Pursuing 3rd year of 3-year diploma after 10th (and continuous education) with NA of experience OR Completed 3 year diploma after 10th OR Pursuing 2nd year of 2-year diploma after 12th (and continuous education) with NA of experience OR 2-year Diploma after 12th grade (in any field) with NA of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 4 and with minimum education as 8th Grade pass) with 1.5 years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/03/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-4.5-IT-00355-2023-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/ITS/N1217: Conceptualize Cloud Security

Description

It focus on the principles and practices of ensuring the security and integrity of data and applications in cloud environments

Scope

The scope covers the following :

- Cloud Security

Elements and Performance Criteria

Cloud Security

To be competent, the user/individual on the job must be able to:

- PC1.** Concepts pertaining to Basic Cloud Security and Cloud Infrastructure Security
- PC2.** Insight into data Security and Storage
- PC3.** Exposure to various Cloud Security considerations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Concepts based on cloud Security

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cloud Security</i>	14	7	5	-
PC1. Concepts pertaining to Basic Cloud Security and Cloud Infrastructure Security	-	-	-	-
PC2. Insight into data Security and Storage	-	-	-	-
PC3. Exposure to various Cloud Security considerations	-	-	-	-
NOS Total	14	7	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1217
NOS Name	Conceptualize Cloud Security
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1216: Conceptualize Docker and Containers

Description

Conceptualize Docker and Containers provides an overview of containerization technology, with a focus on Docker

Scope

The scope covers the following :

- Docker and Containers

Elements and Performance Criteria

Docker and container

To be competent, the user/individual on the job must be able to:

- PC1.** Create and Configure Containers and Dockers.
- PC2.** Introduction to Docker imaging and Configuring the same.
- PC3.** In-depth insight into the configuration of Containers Orchestration.
- PC4.** Adequate exposure to networking concepts used in Dockers

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Docker and Container

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Docker and container</i>	14	7	5	-
PC1. Create and Configure Containers and Dockers.	-	-	-	-
PC2. Introduction to Docker imaging and Configuring the same.	-	-	-	-
PC3. In-depth insight into the configuration of Containers Orchestration.	-	-	-	-
PC4. Adequate exposure to networking concepts used in Dockers	-	-	-	-
NOS Total	14	7	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1216
NOS Name	Conceptualize Docker and Containers
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1215: Emphasizing Cloud Computing Services

Description

It includes importance of cloud-based solutions, emphasizing the benefits of scalability, cost-efficiency, and flexibility for businesses

Scope

The scope covers the following :

- Management & Operational of Cloud Computing

Elements and Performance Criteria

VMware

To be competent, the user/individual on the job must be able to:

PC1. SaaS Cloud Deployment with Open Source Tools

PC2. Introduction to PaaS Cloud and SaaS Cloud Deployment with Google App Engine

PC3. • Knowledge of IaaS Cloud, install and configure ESXi Server along with setting up of
• a Private Cloud With Vsphere

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Choosing adequate cloud Infrastructure to be used for deployment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>VMware</i>	14	6	5	-
PC1. SaaS Cloud Deployment with Open Source Tools	-	-	-	-
PC2. Introduction to PaaS Cloud and SaaS Cloud Deployment with Google App Engine	-	-	-	-
PC3. - Knowledge of IaaS Cloud, install and configure ESXi Server along with setting up of - a Private Cloud With Vsphere	-	-	-	-
NOS Total	14	7	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1215
NOS Name	Emphasizing Cloud Computing Services
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1214: Managing Cloud Architecture with Open Stack

Description

Managing Cloud Architecture with OpenStack involves overseeing and maintaining cloud infrastructure using the open-source OpenStack platform

Scope

The scope covers the following :

- Setup Private Cloud using Open stack

Elements and Performance Criteria

Openstack

To be competent, the user/individual on the job must be able to:

- PC1.** • Basic components and Architectures of open Stack along with Installation of Open stack
- PC2.** • Configure and manage Open Stack Client and Horizon dashboard.
• vice
- PC3.** Configure and manage OpenStack Identity Management
- PC4.** Configure and manage Image management.
- PC5.** Configure and manage Compute Server

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Openstack and its component

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Openstack</i>	16	7	5	-
PC1. • Basic components and Architectures of open Stack along with Installation of Open • stack	-	-	-	-
PC2. • Configure and manage Open Stack Client and Horizon dashboard. • vice	-	-	-	-
PC3. Configure and manage OpenStack Identity Management	-	-	-	-
PC4. Configure and manage Image management.	-	-	-	-
PC5. Configure and manage Compute Server	-	-	-	-
NOS Total	16	7	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1214
NOS Name	Managing Cloud Architecture with Open Stack
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	1.5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1213: Conceptualize Advance Linux

Description

Conceptualize Advanced Linux" is an advanced course designed to deepen students' knowledge and proficiency in the Linux operating system. This course delves into complex Linux concepts, including system administration, security, scripting, and network configurations.

Scope

The scope covers the following :

- Linux Administration

Elements and Performance Criteria

Linux Operating System

To be competent, the user/individual on the job must be able to:

PC1. In-depth knowledge of advanced Utilities and Services in Linux

PC2. • Setting up of different services of Linux like DNS Server, Web Server, Mail Server,
• File Server, and Squid Proxy Server

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Linux Command and its kernel

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Linux Operating System</i>	16	7	-	-
PC1. In-depth knowledge of advanced Utilities and Services in Linux	-	-	-	-
PC2. - Setting up of different services of Linux like DNS Server, Web Server, Mail Server, - File Server, and Squid Proxy Server	-	-	-	-
NOS Total	16	7	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1213
NOS Name	Conceptualize Advance Linux
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1212: Implementing Virtualization and Basics of Cloud Computing

Description

It includes principles and practical aspects of virtualization and cloud technology. Students learn to create and manage virtualized environments, optimize resource allocation, and understand key cloud computing concepts. This course equips individuals with the knowledge and skills needed to work with virtualization technologies and explore the foundational elements of cloud computing, making them well-prepared for roles in IT and cloud-related fields

Scope

The scope covers the following :

- Virtualization and Cloud Computing

Elements and Performance Criteria

Virtualization

To be competent, the user/individual on the job must be able to:

- PC1.** Differentiating between Open source Virtualization and proprietary Virtualization
- PC2.** Installation and use Open source virtualization tools - KVM, Hyper-V and Xen
- PC3.** Creation of Virtual Machine using Virtualization Tools
- PC4.** Cloud Computing Concepts and would be able to differentiate Cloud computing vs. Cluster computing vs. Grid computing.
- PC5.** Conceptualize Cloud Characteristics and Working of Cloud Computing along with Features Of Cloud Computing
- PC6.**
 - Knowledge pertaining to Types of Cloud , Enlightened about the Benefits of Cloud Computing and Applications of Cloud
 - Computing In industries ,In-depth coverage of Cloud Platform ranging from its benefits to its implementation
 - and its scope and its pros and cons

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Differentiate and benefit of Open source Virtualization and proprietary Virtualization.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Virtualization</i>	58	28	10	-
PC1. Differentiating between Open source Virtualization and proprietary Virtualization	-	-	-	-
PC2. Installation and use Open source virtualization tools - KVM, Hyper-V and Xen	-	-	-	-
PC3. Creation of Virtual Machine using Virtualization Tools	-	-	-	-
PC4. Cloud Computing Concepts and would be able to differentiate Cloud computing vs. Cluster computing vs. Grid computing.	-	-	-	-
PC5. Conceptualize Cloud Characteristics and Working of Cloud Computing along with Features Of Cloud Computing	-	-	-	-
PC6. • Knowledge pertaining to Types of Cloud , Enlightened about the Benefits of Cloud Computing and Applications of Cloud • Computing In industries ,In-depth coverage of Cloud Platform ranging from its benefits to its implementation • and its scope and its pros and cons	-	-	-	-
NOS Total	58	28	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1212
NOS Name	Implementing Virtualization and Basics of Cloud Computing
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1211: Introduction to Network Management

Description

It includes fundamental topics of network monitoring, device configuration, performance optimization, and troubleshooting. Students gain insights into the essential tools and techniques used in maintaining and ensuring the efficiency and security of computer networks.

Scope

The scope covers the following :

- Networking

Elements and Performance Criteria

Networking

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks
- PC2.** Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topology used in networking.
- PC3.**
 - Explain various Transmission modes, Ethernet and the Classification Of
 - Transmission Media.
- PC4.**
 - In-depth idea about various networking devices like Repeaters, Hubs, Bridges,Switches and routers
 -
- PC5.** Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols.
- PC6.** IP addressing, Classes, IP Add

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Classification Of Transmission Media and will have in depth idea about various networking devices and Conceptualization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Networking</i>	34	12	-	-
PC1. Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks	-	-	-	-
PC2. Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topology used in networking.	-	-	-	-
PC3. • Explain various Transmission modes, Ethernet and the Classification Of • Transmission Media.	-	-	-	-
PC4. • In-depth idea about various networking devices like Repeaters, Hubs, Bridges,Switches and routers •	-	-	-	-
PC5. Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols.	-	-	-	-
PC6. IP addressing, Classes, IP Add	-	-	-	-
NOS Total	34	13	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1211
NOS Name	Introduction to Network Management
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1210: Conceptualize Operating Systems (Windows and Linux)

Description

This module focuses on developing the foundational knowledge and understanding of operating systems, with a specific emphasis on Windows and Linux environments. It covers core concepts, features, and functionalities of these operating systems, including file systems, user management, process management, and system configurations. Students will learn how to conceptualize and navigate these operating systems, gaining essential skills to work effectively with Windows and Linux in an IT environment. The module includes both theoretical and practical components, allowing students to apply their knowledge to real-world operating system scenarios

Scope

The scope covers the following :

- Able to understand Operating System

Elements and Performance Criteria

Operating System

To be competent, the user/individual on the job must be able to:

- PC1.** Configure Network Settings and perform user and group management
- PC2.** Knowledge of various management service roles and get acquainted with the concept of remote desktop management
- PC3.** Installation of Windows and Linux Operating System
- PC4.** Creation of disk partitions and configure Active directory.
- PC5.**
 - Conceptualisation of Group policy, work in Power Shell and run various Windows commands
- PC6.** Basic Commands in Linux and implement administrator tasks with sudo.
- PC7.** Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Gaining Knowledge of how to run and configure several Linux services.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operating System</i>	34	14	-	-
PC1. Configure Network Settings and perform user and group management	-	-	-	-
PC2. Knowledge of various management service roles and get acquainted with the concept of remote desktop management	-	-	-	-
PC3. Installation of Windows and Linux Operating System	-	-	-	-
PC4. Creation of disk partitions and configure Active directory.	-	-	-	-
PC5. • Conceptualisation of various Group policy, work in PowerShell and run various Windows commands	-	-	-	-
PC6. Basic Commands in Linux and implement administrator tasks with sudo.	-	-	-	-
PC7. Acquire skills pertaining to group user management, Piping and Grouping and also mounting of disk	-	-	-	-
NOS Total	34	14	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1210
NOS Name	Conceptualize Operating Systems (Windows and Linux)
Sector	Information Technology Sector
Sub-Sector	Cloud Computing & Virtualization
Occupation	Cloud Management
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1217.Conceptualize Cloud Security	14	7	5	0	26	7
NIE/ITS/N1216.Conceptualize Docker and Containers	14	7	5	0	26	7
NIE/ITS/N1215.Emphasizing Cloud Computing Services	14	7	5	0	26	8
NIE/ITS/N1214.Managing Cloud Architecture with Open Stack	16	7	5	0	28	8
NIE/ITS/N1213.Conceptualize Advance Linux	16	7	0	0	23	7
NIE/ITS/N1212.Implementing Virtualization and Basics of Cloud Computing	58	28	10	0	96	28
NIE/ITS/N1211.Introduction to Network Management	34	13	0	0	47	13

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1210.Conceptualize Operating Systems (Windows and Linux)	34	14	0	0	48	13
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	220	120	30	-	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
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Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.

Qualification Pack

National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (K)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (G)	Core skills or Generic Skills are a group of skills that are the key to learning and working in world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.